

Case Study of Sea Breeze Propagation across I-90 TRACON

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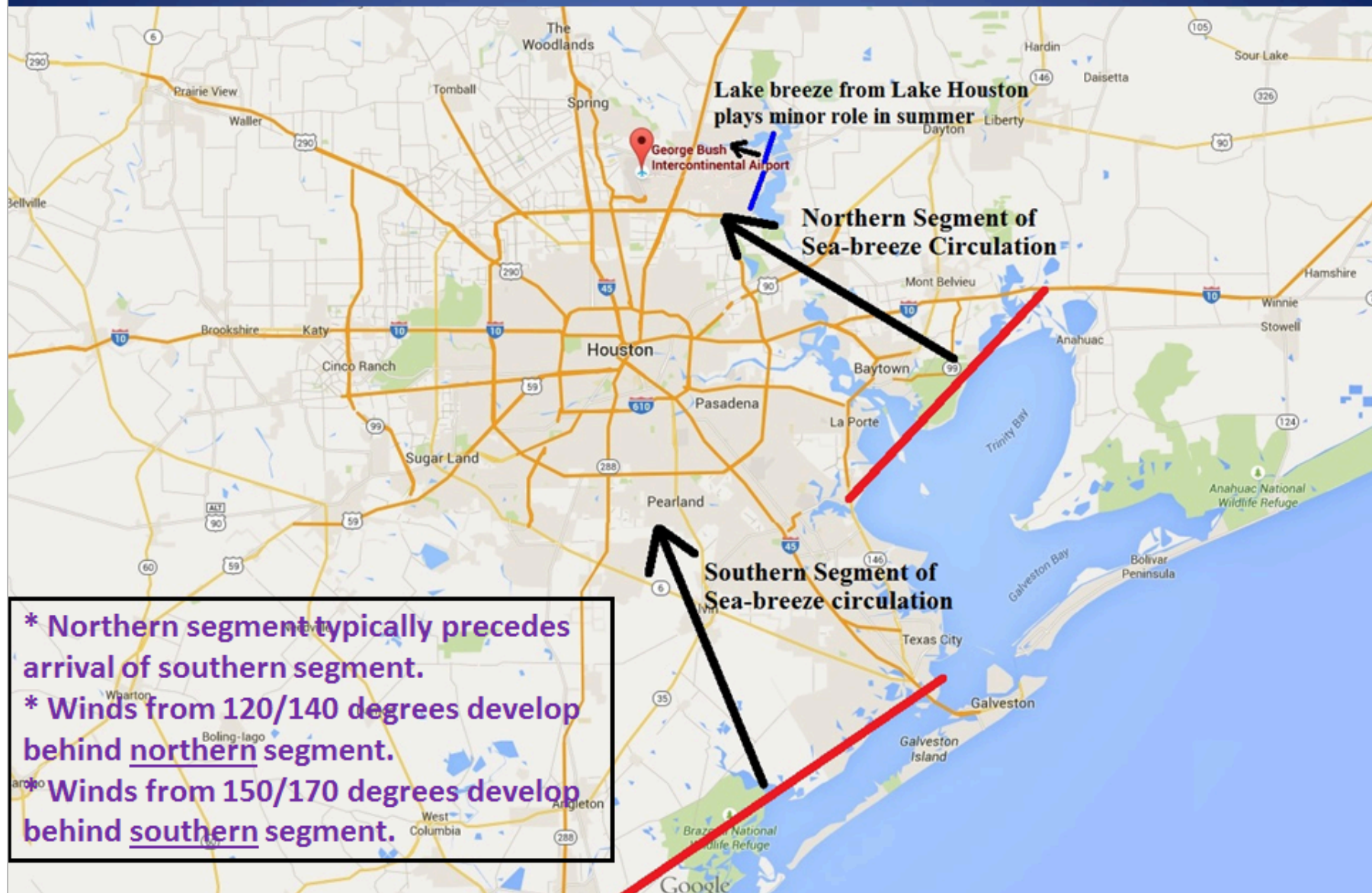


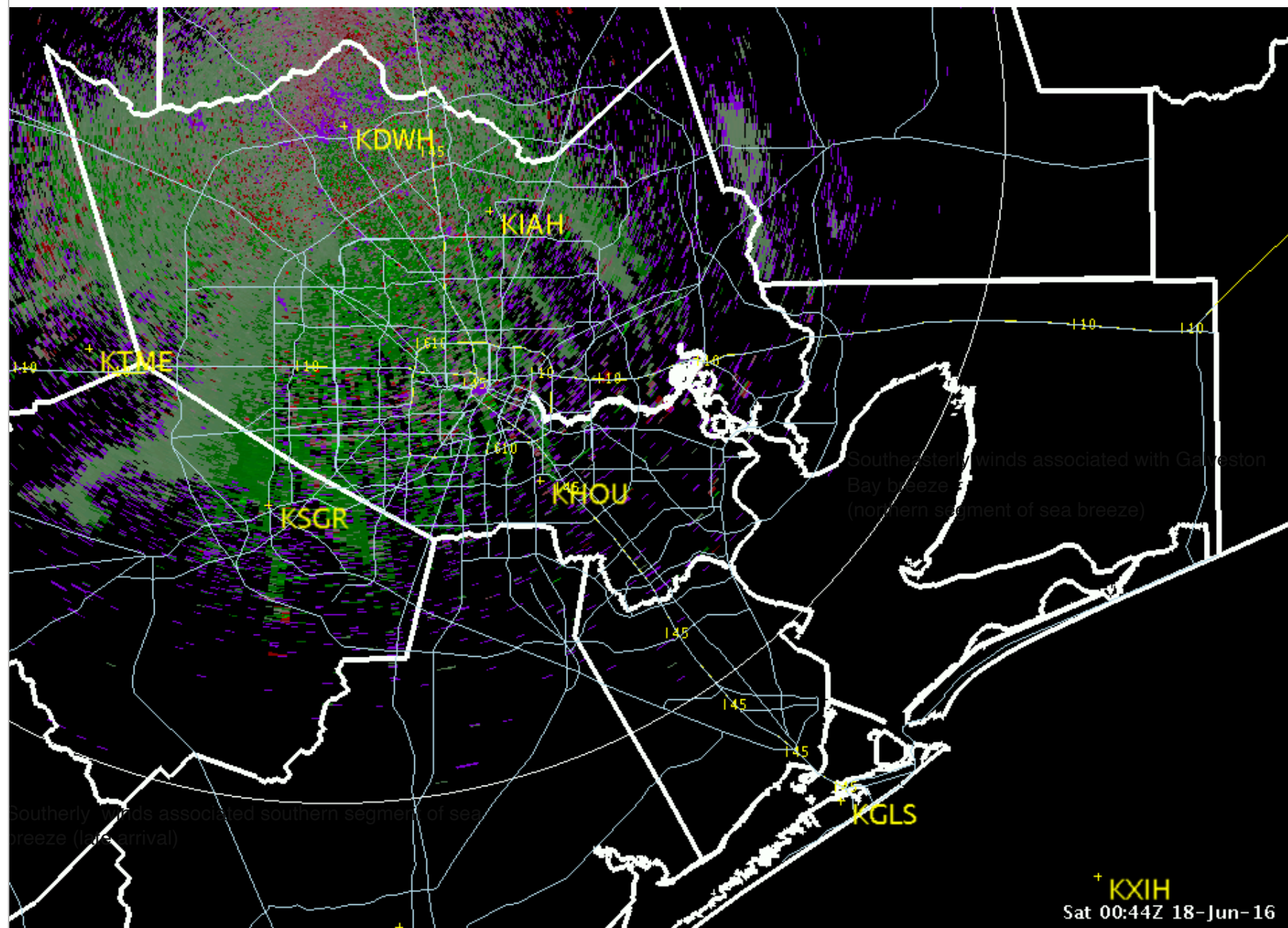
This tutorial is follow-up to presentation titled Forecasting Summertime Winds at KIAH

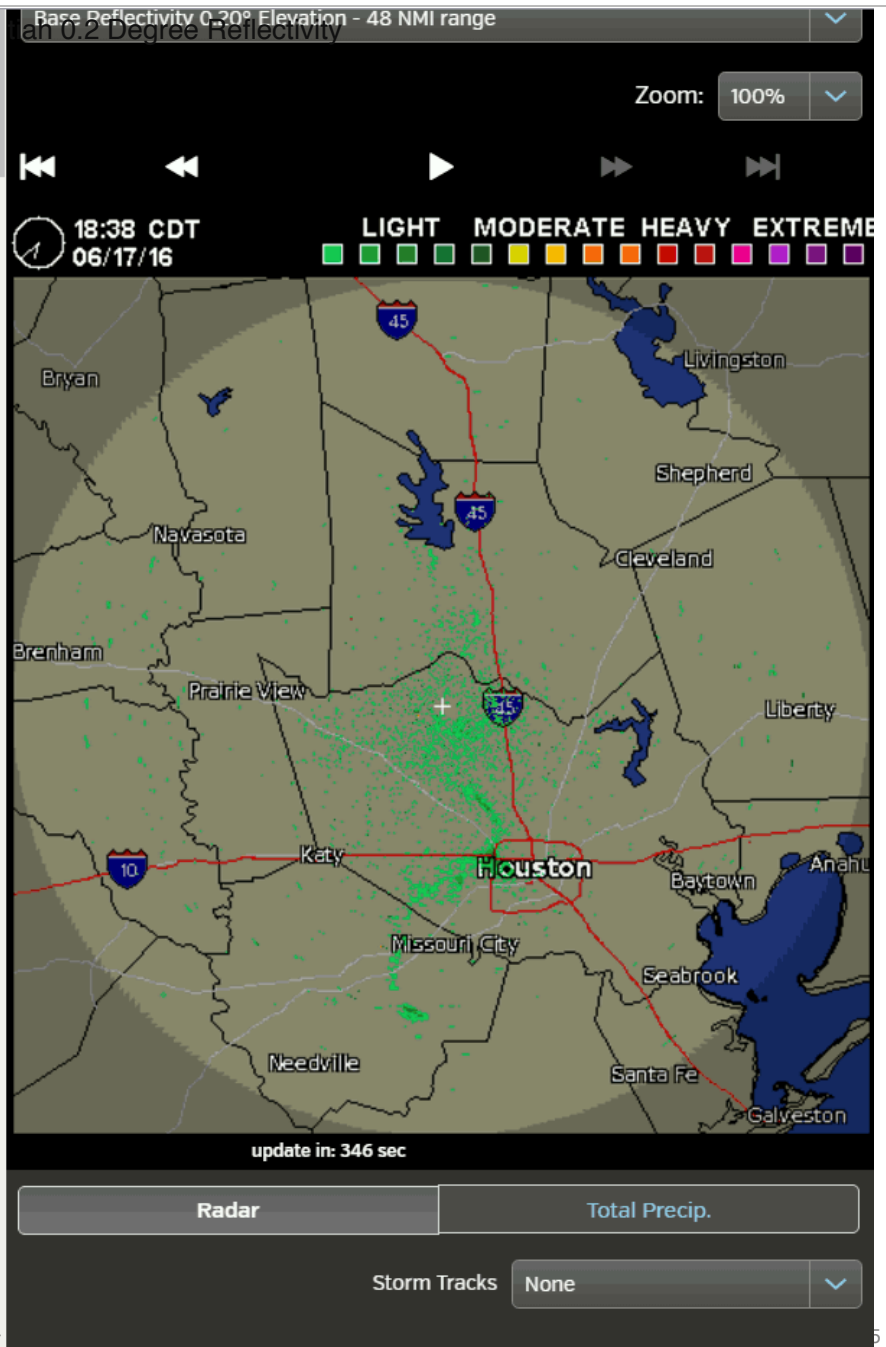
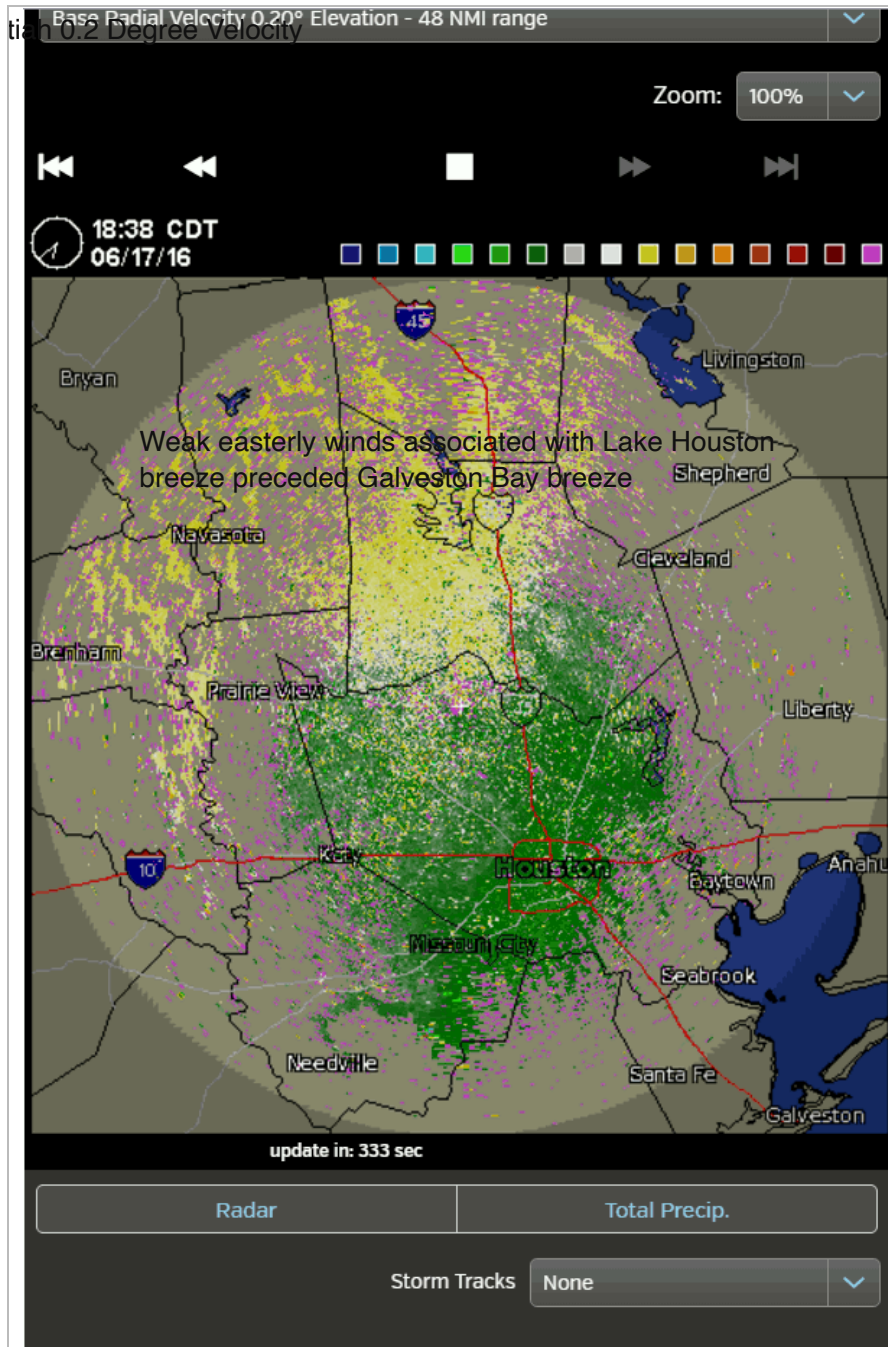
The event took place during afternoon of June 17, 2016.

The following slides contain radar loops of tiah radar.

Geography of Houston Metro Area

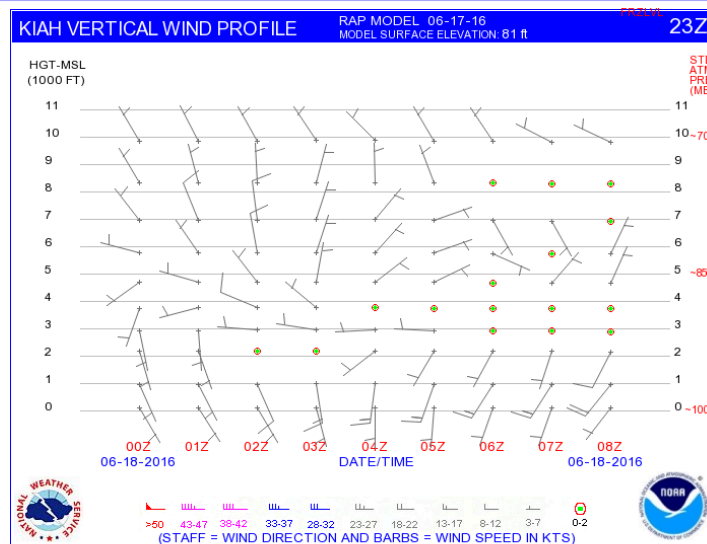
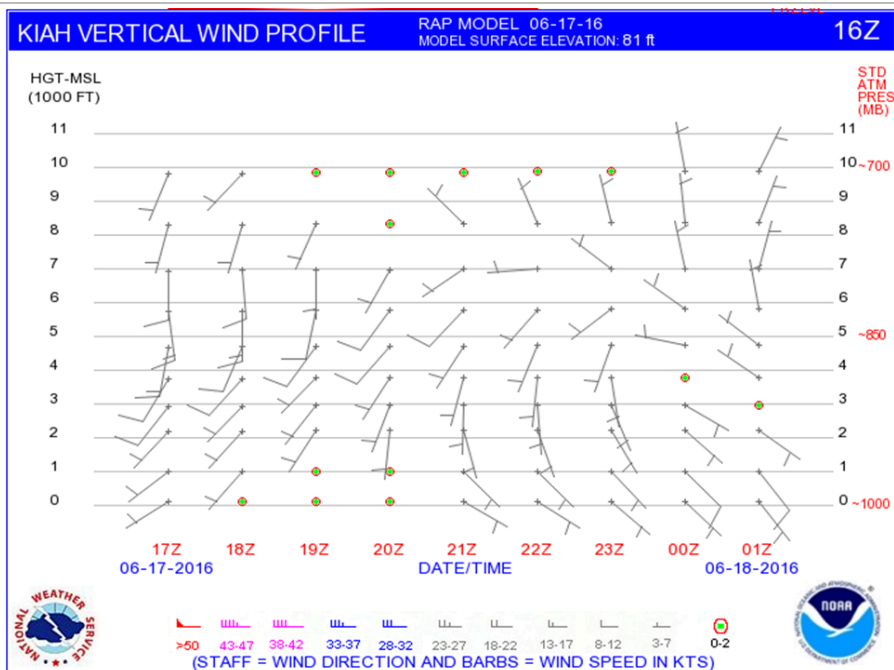






KIAH RAP Vertical Wind
Profile - June 17, 2015
16Z issuance - top image
23Z issuance – bottom

NOTE: 17Z issuance indicated light SE through southwest winds (i.e., 5 knots) would extend from surface through the lowest 5,000 feet and last through at least 01Z.



This graphic displays forecast vertical wind profiles over the selected location from the surface to 10,000 feet AGL. Forecast winds are displayed at 1-hour intervals out through 9 hours. The graphic is updated hourly based on the latest run of the RAP model. Wind speeds are color-coded with speeds 28-37 kts displayed in blue, 38-47 kts in pink, and winds greater than 47 kts in red. Wind direction is indicated by the staff extending from the crosshair using a 360 degree compass, where 180 degrees indicates a south wind and 360 degrees a north wind. The solid red line indicates the highest forecast freezing level for the given time.

What Can We Learn?

- KIAH RAP Vertical Wind Profile graphics can be a valuable tool in forecasting afternoon winds in the summer.
 - ➡ Identify potential changes in wind direction/speed.
 - ➡ Light winds within lowest 5,000 feet will favor arrival of northern segment of sea breeze with winds between 120/140 degrees.
 - ➡ Southerly winds of 10 knots or stronger within lowest 5,000 feet will preclude arrival of northern segment with winds backing no more than 160 degrees.
- Northern segment of sea breeze (if conditions support) normally arrives after 5 PM CDT and precedes arrival of southern segment. However, east or southeast winds in the lowest 5,000 feet will favor an earlier arrival of the northern segment.
- Inflow into thunderstorm development near KIAH can result in both an earlier arrival and prolonged event of easterly winds at KIAH!!

What Can We Learn?

Lake Houston breeze plays a minor role on winds at KIAH in the summertime.

Wind speeds are usually less than 10 knots.
However, easterly winds at KIAH may begin as early as 4 PM.